FERA ফেরা

D4 / DESIGN · VERSION 2.0

UI & UX Design Brief

Bangla interface, component behaviour and accessibility

Specification handover · Implementation unverified

Authors

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Issued 11 October 2026 · Asia/Dhaka (UTC+6)

Invented synthetic-data prototype · Interactive team edition

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> Click a section or use the PDF bookmarks. Requirement IDs link to their definitions. External sources are clickable. The meeting workbook contains blank fillable review fields. Application acceptance and pilot evidence remain pending.

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D4 / DESIGN / V2.0

UI & UX Design Brief

Bangla interface, component behaviour and accessibility · Authors: Sadman Shaharier and Obidit Islam

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Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier (design shared across team)
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Companion	`tokens_v2.css` — byte-identical copy of §16
Inputs	D1, D2 Parts A/B and D3 V2.0; tokens_v2.css; validation report.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.
- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 corrects border contrast, narrow officer layouts, shortcut controls, report-count labels and provider/ referral wording. ASCII sketches are replaced by structured layout specifications. The CSS companion is the exact embedded source.

1 Design objective

A frightened parent on a mid-range Android phone completes a reviewed Bangla report in ≤ 5 min (GOAL-001). An officer at a desk decides a match on evidence in ≤ 2 min (PERSONA-002). Nothing on screen sensationalises a child.

2 Principles

#	Principle	Test
1	Bangla first, plain words	Every parent string reads at a primary-school level; no legal terms without a gloss
2	One thing at a time	Parent screens show one question or one decision per view
3	Show what the AI understood	Every extracted field is visible and editable before submit
4	Calm, not alarming	No red banners, countdowns or flashing on parent screens
5	Evidence before decision	Officer never sees confirm without both records side by side
6	Honest labels	"খসড়া" (draft), "সিনথেটিক" (synthetic), "ক্যাশ" (cached), [MOCK] always visible
7	Status never by colour alone	Icon + text + colour on every status

3 Brand personality

Calm, trustworthy, urgent but not alarming. Voice of a steady neighbour who knows the process. Never: dramatic imagery, child photos, "runaway" language, exclamation-mark urgency, alert-style sirens.

4 User contexts

Context	Parent (PERSONA-001)	Officer (PERSONA-002)	Staff (PERSONA-003)
Device	Android phone, 360–412 px wide, 4G, often one hand	Laptop/desktop, 1280–1920 px, mouse + keyboard	Phone or shared desk PC
State	Distressed, time-pressured, may be outdoors	Focused, many cases, accountable	Busy, interrupted

Context	Parent (PERSONA-001)	Officer (PERSONA-002)	Staff (PERSONA-003)
Need	Reassurance, few steps, voice	Density, comparison, keyboard speed	Short form, quick save

5 Information architecture

Area	Screens
Parent (no account)	Welcome (SCREEN-001) → Intake (SCREEN-002) → Review (SCREEN-003) → Outputs (SCREEN-004); Manual form (SCREEN-005); Status (SCREEN-006)
Staff	Sign-in (SCREEN-008) → Found-child log (SCREEN-007); Disaster mode (SCREEN-015)
Officer	Queue (SCREEN-009) · Match review (SCREEN-010) · Map (SCREEN-011) · Case detail (SCREEN-012) · Duplicates (SCREEN-014)
Admin	Data tools (SCREEN-013); aggregate map/table and metadata-only case detail
Shared	Not found (SCREEN-016)

Officer top nav: "যাচাই তালিকা" / Review queue · "মানচিত্র" / Map · "সাইন আউট" / Sign out.

6 Visual direction

6.1 Palette (light)

Token	Hex	Use	Contrast
`--color-bg`	#FAFAF7	Page	text 15.20:1
`--color-surface`	#FFFFFF	Cards	text 15.90:1
`--color-surface-2`	#F1F3F2	Chat bubbles, table stripes	—
`--color-text`	#1B2421	Body text	15.90:1 on surface
`--color-text-muted`	#4E5A55	Secondary text	7.20:1 on surface; 6.88:1 on bg
`--color-border`	#6F7D77	Input borders	4.31:1 on surface; minimum 3.86:1 across all three surfaces
`--color-primary`	#0F5C55	Buttons, links (deep teal)	7.83:1 on surface
`--color-on-primary`	#FFFFFF	Text on primary	7.83:1
`--color-focus`	#2457C5	Focus ring	6.47:1 on surface
`--status-open-fg` / `-bg`	#7A4F00 / #FFF3D6	Open (amber)	6.46:1

Token	Hex	Use	Contrast
`--status-matched-fg` / `-bg`	#1E4E9C / #E3EDFC	Matched (blue)	6.78:1
`--status-closed-fg` / `-bg`	#1E6B3A / #E2F3E8	Closed (green)	5.66:1
`--color-danger-fg` / `-bg`	#B42318 / #FDECEA	Officer-side destructive and field errors only	5.75:1

6.2 Palette (dark)

Token	Hex	Contrast
`--color-bg`	#111816	text 15.32:1
`--color-surface`	#18211E	text 14.01:1
`--color-surface-2`	#212C28	—
`--color-text`	#E8EEEB	14.01:1 on surface
`--color-text-muted`	#A9B5B0	7.78:1 on surface
`--color-border`	#82938B	5.09:1 on surface; minimum 4.46:1 across all three surfaces
`--color-primary`	#5FC4B8	7.90:1 on surface
`--color-on-primary`	#0B1F1C	8.21:1 on primary
`--color-focus`	#8AB4F8	7.81:1 on surface
Open fg / bg	#F2C14E / #33290F	8.54:1
Matched fg / bg	#8DB4F5 / #172640	7.19:1
Closed fg / bg	#7FD39A / #14301F	7.94:1
Danger fg / bg	#F28B82 / #3A1714	6.70:1

V2 ratios are recalculated with the WCAG 2.x relative-luminance formula on 11 Oct 2026. These verify the listed colour pairs; full rendered accessibility, focus, zoom, reflow and screen-reader tests remain pending.

6.3 Typography

Item	Value
Bangla	Noto Sans Bengali (Google Fonts), weights 400, 600
Latin pair	Noto Sans (Google Fonts), weights 400, 600 — matched x-height and stroke with the Bangla face
Loading	Self-hosted woff2 subsets, `font-display: swap`; Bangla subset preloaded on parent routes

Item	Value
Scale	14 / 16 / 18 / 22 / 28 px (`--font-size-sm` … `--font-size-2xl`)
Parent body	18 px, line-height 1.7 (Bangla conjuncts and matras need room)
Officer body	16 px, line-height 1.55
Numerals	Bangla numerals on parent screens; officer screens show both where space allows ("১৩ / 13")

6.4 Spacing, radius, elevation, icons, illustration

Item	Value
Spacing	4-px base: 4, 8, 12, 16, 24, 32, 48, 64
Radius	6 px inputs, 10 px cards, 16 px sheets, 999 px pills
Elevation	Level 1 cards: `0 1px 2px rgba(0,0,0,.08)` ; level 2 dialogs: `0 8px 24px rgba(0,0,0,.16)`
Icons	Outline, 1.75 px stroke, 20/24 px (Lucide set); always paired with text on actions and statuses
Status icons	Open: circle-dot; Matched: git-compare; Closed: circle-check; Referral recorded: hand-heart with simulation text
Illustration	Simple flat line drawings of places (house, station, map pin). No children's faces, no stock photos, no silhouettes of lone children

7 Layout and responsive behaviour

Breakpoint	Layout
< 600 px	Parent screens single column, max width 480 px, sticky bottom action bar (safe-area aware)
600–1023 px	Staff form two-column for structured fields; officer screens stack columns
≥ 1024 px	Officer: 12-column grid, max 1280 px; match review two equal columns + 320 px decision panel; map 2/3 + table 1/3

Text scales to 200 % without horizontal scroll on parent screens. Tap targets ≥ 44 × 44 px.

8 Component inventory

ID	Component	Used on	Key states
COMP-022	App header with language toggle	All	bn / en
COMP-023	Synthetic badge "সিনথেটিক"	Any seeded record	—

ID	Component	Used on	Key states
COMP-024	Chat bubble (assistant / user)	SCREEN-002	sending, sent, error
COMP-025	Voice record button with timer	SCREEN-002, SCREEN-007	idle, requesting, recording, uploading, disabled
COMP-026	Transcript editor	SCREEN-002	editable, saved
COMP-027	Question prompt with quick-reply chips	SCREEN-002	—
COMP-028	Editable field card	SCREEN-003, SCREEN-007	filled, unknown, error, edited
COMP-029	Confirm bar (checkbox + submit)	SCREEN-003, SCREEN-007	disabled until checked, submitting
COMP-030	Output card with copy	SCREEN-004	loading, ready, copied, limit reached
COMP-031	Poster preview + PNG download	SCREEN-004	rendering, ready
COMP-032	Manual case form	SCREEN-005	prefilled, error
COMP-033	Status badge (icon + text + colour)	SCREEN-004, -006, -009, -011, -012	open, matched, closed, referred
COMP-034	Found-child form	SCREEN-007, SCREEN-015	—
COMP-035	Review queue list	SCREEN-009	loading, empty, populated
COMP-036	Candidate row (rank, score, reason)	SCREEN-009	focused, opened
COMP-037	Side-by-side evidence table	SCREEN-010, SCREEN-014	agree, differ, unknown per row
COMP-038	Decision panel (confirm / reject + reason)	SCREEN-010	idle, confirming dialog, submitting
COMP-039	Close-case dialog	SCREEN-012	outcome chosen, note entered
COMP-040	District map	SCREEN-011	tiles, no-tiles fallback
COMP-041	District table	SCREEN-011	sorted by open count
COMP-042	Audit timeline	SCREEN-012	—
COMP-043	Admin action panel	SCREEN-013	typed confirmation

ID	Component	Used on	Key states
COMP-044	Inline alert / toast	All	info, success, error
COMP-045	Empty state	SCREEN-006, -009, -013, -016	—
COMP-046	Loading state with 10 s message	SCREEN-002, -004, -007	< 10 s, ≥ 10 s
COMP-047	Sign-in form	**SCREEN-008**	error, rate-limited
COMP-048	Cached badge "ক্যাশ"	Any AI output from demo cache	—
COMP-049	Duplicate banner "সম্ভাব্য একই শিশু"	SCREEN-010, SCREEN-012	—
COMP-050	Consent notice	**SCREEN-001**	—

9 Interaction guidelines

Interaction	Rule
Voice record	Tap to start, tap to stop (no press-and-hold). Live timer "০:৩৮ / ২:০০". Auto-stop at 120 s. Haptic tick on start/stop where supported. After stop: "পাঠাচ্ছি..." / Sending.... Transcript appears in an editable box with "ঠিক আছে, পাঠান" / OK, send.
Review before register/ export; provider notice earlier	Cards in the order the parent spoke. Unknown fields read "অজানা — যোগ করুন" / Unknown — add. Required-field errors are visible; submit stays disabled until "আমি যাচাই করেছি" / I have checked is ticked.
Side-by-side evidence	Missing on the left, found on the right, same field order. Row verdict icon + word. Reason line and score on top; score labelled "মিলের মাত্রা ৮৭/১০০" / Similarity 87/100, never "%" or "probability".
Confirm / reject	Confirm opens a dialog restating the referral rule; focus starts on "ফিরে যান" / Go back. Reject requires a reason (min 3 chars) and quick chips: "বয়স মেলে না" / Age differs, "জেলা দূরে" / District far, "পোশাক আলাদা" / Clothing differs.
Copy and download	Copy shows "কপি হয়েছে" for 2 s and announces it. Poster PNG 1080 × 1350 px, white background, Bangla text, 13219 and the explicitly chosen contact option, "খসড়া" footer.
Keyboard (officer)	`J/K` or ↑/↓ move in queue; `Enter` open; `C` confirm (opens dialog); `R` reject; disabled while typing and switchable/remappable; `Esc` close dialog. Shortcuts listed under "?"

10 Microcopy guidelines

Rule	Bangla / English example
Address the parent with respect (আপনি)	"আপনার শিশুর কথা বলুন" / Tell us about your child

Rule	Bangla / English example
Say what happens next	"জমা দেওয়ার পর জিডি খসড়া ও পোস্টার পাবেন" / After submitting you'll get a GD draft and poster
Never promise outcomes	Not "আমরা খুঁজে দেব" / We will find. Use "মিল খোঁজা হবে" / Matches will be checked
State Fera's limits	"ফেরা পুলিশ বা সিআইডি-কে সতর্কবার্তা পাঠায় না। জরুরি হলে ১৩২১৯-এ কল করুন।" / Fera does not alert police or CID. In an emergency, call 13219.
Neutral words about the child	"শেষবার দেখা গিয়েছিল" / Last seen; never "পালিয়ে গেছে" / ran away
Errors: what happened + what to do	"কথাগুলো পরিষ্কার শোনা যায়নি। আবার বলুন বা লিখে দিন।" / We couldn't hear that clearly. Try again or type.
Officer copy may be denser	"৩টি সম্ভাব্য মিল · সর্বোচ্চ ৮৭" / 3 likely matches · top 87

11 Forms and validation

Field	Input	Validation	Message (Bangla / English)
Age	Numeric keypad; accepts ০–৯ and 0–9; range option	0–17 (missing)	"বয়স ০ থেকে ১৭-এর মধ্যে দিন" / Enter an age from 0 to 17
Sex	4 large buttons	Required	"একটি বেছে নিন" / Choose one
District	Searchable list, Bangla + English names, aliases ("চট্টগ্রাম / Chattogram / Chittagong")	One of 64	"তালিকা থেকে জেলা বেছে নিন" / Pick a district from the list
Last seen	Date + part-of-day chips (সকাল, দুপুর, বিকেল, সন্ধ্যা, রাত) or exact time	Not in future	"ভবিষ্যতের সময় দেওয়া যাবে না" / The time can't be in the future
Phone	`inputmode="tel"`; Bangla digits accepted	`^01[0-9]{9}$` after conversion	"০১ দিয়ে শুরু ১১ সংখ্যার নম্বর দিন" / Enter an 11-digit number starting 01
Free text	Auto-growing textarea	20–2,000 chars	"আরও একটু বিস্তারিত লিখুন" / Please add a little more detail

Validate on blur and on submit, never on each keystroke. Errors appear under the field, linked by `aria-describedby`, plus a summary at the top on submit.

12 Loading, empty, success and error states

State	Pattern
Loading < 300 ms	Nothing shown
Loading 300 ms – 10 s	COMP-046 spinner + verb ("বুঝে নিচ্ছি..." / Understanding...)

State	Pattern
Loading ≥ 10 s	"এখনও কাজ চলছে, একটু অপেক্ষা করুন" / Still working, please wait + "ফর্মে লিখুন" / Use the form link
Empty	COMP-045 line illustration + one sentence + one action
Success	Inline confirmation; parent screens use teal check, never confetti
Error	COMP-044 with message from D2 Part B §13; parent errors use neutral surface with icon, not red fill

13 Accessibility (WCAG 2.2 AA)

Requirement	Implementation
Keyboard access	All actions reachable by Tab; officer shortcuts (§9); dialogs trap focus and restore it
Visible focus	3 px `--color-focus` outline, 2 px offset; never hidden by sticky bars (2.4.11)
Contrast	Text ≥ 4.5:1; Control-identifying boundaries ≥ 3:1 against every adjacent surface; decorative separators excluded (§6)
Labels	Every input has a visible label; icons have `aria-label` in the current language
Alt text	Illustrations decorative (`alt=""`); poster PNG has a text equivalent shown beside it
Captions / transcripts	Every voice note has its editable transcript; audio is never the only record
Clear errors	Field + summary; error text says how to fix
Target size	≥ 44 × 44 px (exceeds 2.5.8 minimum of 24 px)
Responsive text	320 CSS px reflow, 200 % text zoom and 400 % viewport zoom without loss; `rem` units
No colour-only status	COMP-033 always icon + text + colour
Language	`lang="bn"` / `lang="en"` set per element for screen readers
Redundant entry	Draft reused across S-002 → S-003 → S-005; parent never retypes (3.3.7)
Motion	No auto-animation; respects `prefers-reduced-motion`

14 Screen-by-screen descriptions

Screen	Description
SCREEN-001	Logo wordmark "ফেরা", 2-line explanation, consent notice, Start button, small `tel:13219` link.
SCREEN-002	Chat column; assistant bubble left, parent right; large round mic button above a text box; transcript box after recording; one question at a time with quick-reply chips.

Screen	Description
SCREEN-003	Title "যা বুঝেছি" / What we understood; stacked field cards with edit icons; "অজানা" chips; checkbox + submit in sticky bar.
SCREEN-004	Reference "M-7K3Q2A" in large mono digits; status badge "খোলা"; three output cards; poster preview; 13219 call link.
SCREEN-005	Plain one-column form with the same fields as SCREEN-003; banner explains the assistant was unavailable.
SCREEN-006	Reference, status badge, last updated time, "তথ্য সংশোধন" button when editable.
SCREEN-007	Free-text box + structured fields side by side on desktop; save card shows ref "F-..." and "মিল খোঁজা হয়েছে".
SCREEN-008	Centered form, role named in title ("কর্মকর্তা সাইন ইন" / Officer sign-in).
SCREEN-009	Table: rank-1 score, missing ref, child summary, district, candidate count, newest time; filter by district.
SCREEN-010	Header with score, reason, duplicate banner; two columns of fields with verdict icons; right decision panel.
SCREEN-011	Map with district markers sized by total and labelled with three counts; adjacent table; totals strip; synthetic toggle.
SCREEN-012	Case fields, status badge, linked duplicates, referral card, close button, audit timeline.
SCREEN-013	Three panels: verify officers [MOCK], redact identifying payloads (typed confirm), import (Should).
SCREEN-014	Two same-kind records side by side; "একই শিশু" / "আলাদা" buttons.
SCREEN-015	Found-child form with "পরিকল্পিত সম্প্রসারণ" banner.
SCREEN-016	Line illustration, "তথ্যটি পাওয়া যায়নি", home link.

15 Layout specifications

These replace ASCII wireframes with precise responsive regions. PDF diagrams are vector-rendered; the source diagrams remain editable.

Screen/ layout	Regions in reading order	Behaviour
Parent intake	Header/language; processing notice; synthetic badge; conversation; editable transcript; text/voice controls; manual fallback	One column at 320px, voice never sole input; live status announced
Parent review	Required-field summary; child; event interval/place/ district; reporter contact; accuracy confirmation; submit	Known values preserved; unknown mandatory fields linked to manual completion

Screen/ layout	Regions in reading order	Behaviour
Submitted/ output	Reference and matching state; GD text; poster preview/ contact choice; share acknowledgement; download; clear-device	Fixed draft label; no police filing or alert claim
Found entry	Role and synthetic cue; description; optional age; place/ district/time/contact; review/save	Own found-only intake; no candidate payload in response
Officer queue	Role; district filter; missing/found report basis; current qualified candidates; empty/failed state	Tab navigation plus optional focus-scoped shortcuts; no auto hidden duplicates
Candidate review	Similarity label; versions; side-by-side supported values; unknown/contradiction rows; checklist; confirm/reject	Stack columns on narrow viewport; reason required on reject; score not identity proof
Decision dialog	Simulated-referral warning; reviewer checklist; Go back first; explicit decision	Focus trapped/restored; no keyboard shortcut triggers in note fields
Dashboard	Missing report counts; found report counts; confirmed pairs; district outlines; table; officer drill-through	Table alternative, zero districts, exact counting labels; no unique-child/recovery implication
Admin	Metadata; mock user list/verify; eligible redaction summary and typed confirmation	No child/candidate detail; external-copy redaction limitation stated

15.1 Required V2 microcopy

Moment	Bangla	English
Demo banner	সিনথেটিক তথ্য · কর্মকর্তা যাচাই সিমুলেশন	Synthetic data · officer verification simulated
Provider notice	কথা বা লেখা বোঝার জন্য বাইরের এআই সেবা ব্যবহার হতে পারে। এই সংস্করণে বাস্তব শিশুর তথ্য দেবেন না।	External AI may process audio/text. Do not enter real child information in this version.
Before submit	জমা দেওয়ার আগে তথ্য যাচাই করুন। ফেরা পুলিশে জিডি জমা দেয় না।	Review before submission. Fera does not file a police GD.
Confirm result	পর্যালোচকের সিদ্ধান্ত নথিভুক্ত। রেফারেল নথিভুক্ত (সিমুলেশন)।	Reviewer decision recorded. Referral recorded (simulation).
Score warning	মিলের মাত্রা পরিচয়ের প্রমাণ নয়।	Similarity is not proof of identity.
Duplicate cue	সম্ভাব্য একই শিশু · আলাদা প্রতিবেদন দৃশ্যমান	Possible duplicate · separate reports remain visible
Poster share	প্রকাশের আগে নম্বর ও স্থানের তথ্য দেখুন। ডাউনলোড করা কপি ফেরা মুছতে পারে না।	Review phone/location before sharing. Fera cannot remove downloaded copies.
Matching failure	প্রতিবেদন জমা হয়েছে। মিল খোঁজা সম্পূর্ণ হয়নি; আবার চেষ্টা করুন।	Report saved. Matching is incomplete; retry.
Cached fixture	ক্যাশ · পূর্বপ্রস্তুত সিনথেটিক উদাহরণ	Cached · prepared synthetic example

Moment	Bangla	English
Closure	সমন্বয়ের রেকর্ড বন্ধ হয়েছে; উদ্ধার নিশ্চিত করা হয়নি।	Coordination record closed; rescue is not established.

16 Design tokens

The exact companion is tokens_v2.css. Control borders have been strengthened against all light/dark surfaces; component usage still needs visual and keyboard checks.

```
/* Fera design tokens V2.0 - D4 section 16. Authors: Sadman Shaharier; Obidit Islam. Light default;
dark via prefers-color-scheme or [data-theme]. */
:root {
  /* Colour — light */
  --color-bg: #FAFAF7;
  --color-surface: #FFFFFF;
  --color-surface-2: #F1F3F2;
  --color-text: #1B2421;
  --color-text-muted: #4E5A55;
  --color-border: #6F7D77; /* ≥ 3:1 against all three light surfaces */
  --color-primary: #0F5C55;
  --color-primary-hover: #0B4843;
  --color-on-primary: #FFFFFF;
  --color-focus: #2457C5;
  --status-open-fg: #7A4F00;
  --status-open-bg: #FFF3D6;
  --status-matched-fg: #1E4E9C;
  --status-matched-bg: #E3EDFC;
  --status-closed-fg: #1E6B3A;
  --status-closed-bg: #E2F3E8;
  --color-danger-fg: #B42318;
  --color-danger-bg: #FDECEA;

  /* Typography */
  --font-bn: "Noto Sans Bengali", "Noto Sans", system-ui, sans-serif;
  --font-latin: "Noto Sans", "Noto Sans Bengali", system-ui, sans-serif;
  --font-mono: ui-monospace, "SFMono-Regular", Menlo, Consolas, monospace;
  --font-size-sm: 0.875rem;   /* 14px */
  --font-size-md: 1rem;       /* 16px */
  --font-size-lg: 1.125rem;   /* 18px — parent body */
  --font-size-xl: 1.375rem;   /* 22px */
  --font-size-2xl: 1.75rem;   /* 28px */
  --font-weight-regular: 400;
  --font-weight-semibold: 600;
  --line-height-bn: 1.7;
  --line-height-officer: 1.55;

  /* Spacing (4px base) */
  --space-1: 0.25rem;
  --space-2: 0.5rem;
  --space-3: 0.75rem;
  --space-4: 1rem;
  --space-6: 1.5rem;
  --space-8: 2rem;
  --space-12: 3rem;
  --space-16: 4rem;
```

```
  /* Radius */
  --radius-input: 6px;
  --radius-card: 10px;
  --radius-sheet: 16px;
  --radius-pill: 999px;

  /* Elevation */
  --shadow-1: 0 1px 2px rgba(0, 0, 0, 0.08);
  --shadow-2: 0 8px 24px rgba(0, 0, 0, 0.16);

  /* Interaction */
  --focus-ring: 0 0 0 2px var(--color-surface), 0 0 0 5px var(--color-focus);
  --tap-target-min: 44px;
  --report-flow-min-width: 320px;
  --duration-fast: 120ms;
  --duration-base: 200ms;

  /* Layout */
  --parent-max-width: 480px;
  --officer-max-width: 1280px;
  --decision-panel-width: 320px;
}

@media (prefers-color-scheme: dark) {
  :root:not([data-theme="light"]) {
    --color-bg: #111816;
    --color-surface: #18211E;
    --color-surface-2: #212C28;
    --color-text: #E8EEEB;
    --color-text-muted: #A9B5B0;
    --color-border: #82938B; /* ≥ 3:1 against all three dark surfaces */
    --color-primary: #5FC4B8;
    --color-primary-hover: #7FD3C9;
    --color-on-primary: #0B1F1C;
    --color-focus: #8AB4F8;
    --status-open-fg: #F2C14E;
    --status-open-bg: #33290F;
    --status-matched-fg: #8DB4F5;
    --status-matched-bg: #172640;
    --status-closed-fg: #7FD39A;
    --status-closed-bg: #14301F;
    --color-danger-fg: #F28B82;
    --color-danger-bg: #3A1714;
    --shadow-1: 0 1px 2px rgba(0, 0, 0, 0.4);
    --shadow-2: 0 8px 24px rgba(0, 0, 0, 0.5);
  }
}

:root[data-theme="dark"] {
  --color-bg: #111816;
  --color-surface: #18211E;
  --color-surface-2: #212C28;
  --color-text: #E8EEEB;
  --color-text-muted: #A9B5B0;
  --color-border: #82938B; /* ≥ 3:1 against all three dark surfaces */
  --color-primary: #5FC4B8;
  --color-primary-hover: #7FD3C9;
  --color-on-primary: #0B1F1C;
```

```
  --color-focus: #8AB4F8;
  --status-open-fg: #F2C14E;
  --status-open-bg: #33290F;
  --status-matched-fg: #8DB4F5;
  --status-matched-bg: #172640;
  --status-closed-fg: #7FD39A;
  --status-closed-bg: #14301F;
  --color-danger-fg: #F28B82;
  --color-danger-bg: #3A1714;
  --shadow-1: 0 1px 2px rgba(0, 0, 0, 0.4);
  --shadow-2: 0 8px 24px rgba(0, 0, 0, 0.5);
}

@media (prefers-reduced-motion: reduce) {
  :root {
    --duration-fast: 0ms;
    --duration-base: 0ms;
  }
}
```

17 Usability risks and mitigations

Risk	Mitigation
Parent is distressed and cannot read long text	One question per view; voice first; 18 px text; no jargon
Parent expects Fera to alert police	Consent notice and output labels state Fera does not alert; `tel:13219` always visible
Parent fears data exposure	Consent notice lists who sees what; matched records officer-only; no photos
Child photo privacy	No photo field anywhere; poster is text only; demo uses synthetic text only
AI misheard details shared wrongly	Mandatory review with "আমি যাচাই করেছি" checkbox
Officer over-trusts the score	Score labelled similarity; evidence rows with verdicts; 2-step confirm; referral reminder
Officer fatigue / mis-click	Confirm dialog focuses "Go back"; reject needs reason
Bangla rendering breaks on low-end phones	Self-hosted Noto Sans Bengali subset; tested on 2 Android phones (TASK-036)
Map unreadable or offline	Table always shown beside map; tile-less fallback
Demo shows cached outputs as live	"ক্যাশ" badge on every cached output

IDs introduced

- COMP-022 App header with language toggle

- COMP-023 Synthetic badge
- COMP-024 Chat bubble
- COMP-025 Voice record button
- COMP-026 Transcript editor
- COMP-027 Question prompt
- COMP-028 Editable field card
- COMP-029 Confirm bar
- COMP-030 Output card
- COMP-031 Poster preview
- COMP-032 Manual case form
- COMP-033 Status badge
- COMP-034 Found-child form
- COMP-035 Review queue list
- COMP-036 Candidate row
- COMP-037 Side-by-side evidence table
- COMP-038 Decision panel
- COMP-039 Close-case dialog
- COMP-040 District map
- COMP-041 District table
- COMP-042 Audit timeline
- COMP-043 Admin action panel
- COMP-044 Inline alert / toast
- COMP-045 Empty state
- COMP-046 Loading state
- COMP-047 Sign-in form
- COMP-048 Cached badge
- COMP-049 Duplicate banner
- COMP-050 Consent notice

IDs referenced

- SCREEN-001 – SCREEN-016 (D1)
- GOAL-001, PERSONA-001 – PERSONA-003 (00)
- TASK-036 (D6)

V2 accessibility and export acceptance

Required: 320 CSS-pixel reflow, 200% text zoom, 400% viewport zoom; keyboard and Bangla screen reader walk; focus not obscured; live/error announcements; shortcut disable/remap or focus scope; no shortcuts while typing. Test actual control backgrounds and all states, not tokens alone. axe is complementary, not a conformance certificate. Parent forms remain usable without maps/audio. Test generated poster after fonts load with long Bangla/unknown fields and a text equivalent. Export remains light and readable regardless of UI theme.

References: WCAG non-text contrast, shortcuts, reflow. No AA conformance claim is made before the implemented product is checked.

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